Consent

Thank you for participating in this study. It will include decision-making tasks followed by some questions about yourself.

Estimated study time: 30 minutes.
Your responses are confidential. Whatever information you convey in this study will be anonymized and will not be traced back to you.

Please read all the instructions and information very carefully.
If you have any questions, you may contact Matan Solomon: ma.solomon@campus.technion.ac.il

HEADS UP
Achieving high performance can qualify you for **bonus** payments of up to **1 pound**.

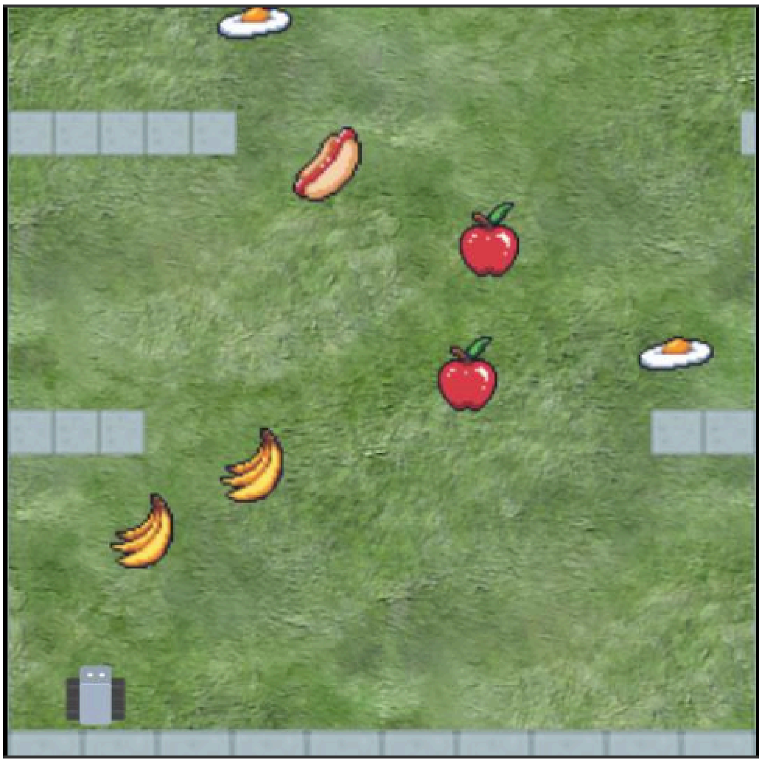
Notice: After choosing one of the options below, click on the blue arrow at the bottom right side of the screen.

I confirm my participation in the study	I DO NOT confirm my participation in the study
○	○

Please record your participant ID in prolific academic:

Part 1: Game instructions

In this study, you will help train an autonomous AI agent to play in the Fruitbot game.
The agent controls a small gray vehicle that collects food items while avoiding walls.
Your goal is to evaluate its performance and give it feedback so it can improve itself.



First, you need to understand the game mechanics.
In the game, the vehicle drives forward automatically. The agent's job is to steer left and right to navigate the scrolling terrain.
Game Rules:
If the vehicle touches a food item, it collects it, and if it touches a wall, the round ends. Touching can be made with any part of the vehicle.

Objectives & Rules:
The goal is to earn the highest score possible by collecting fruits while avoiding obstacles and junk food.

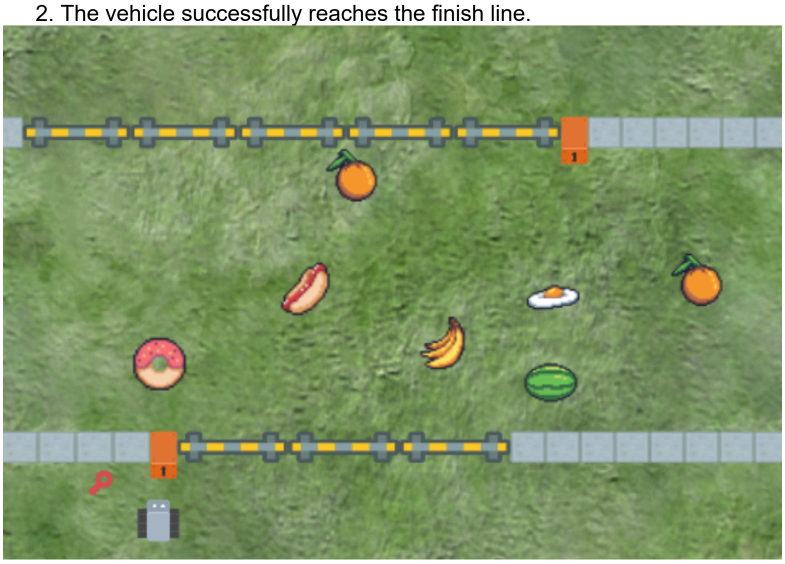
- **Navigation:** The vehicle moves forward constantly. It cannot stop or reverse.
- **Collecting:** If any part of the vehicle touches a food item, it is collected instantly
- **Crashing:** If *any* part of the vehicle (front, sides, or back) grazes a wall, the round ends immediately.
- **Locked Doors:** Some gaps in the walls are blocked by locked doors. The vehicle can "Throw a Key" at the lock to open the gate and pass through. The vehicle has an unlimited number of keys, but it can only throw a new key after the previous one disappears (door example in the image below).

Scoring Guide

- **+1 points (Collect a Fruit):** Apple, Orange, Cherry, Banana, Watermelon, Pineapple.
- **-1 points (Collect a Junk Food):** Popcorn, Ice Cream, Egg, Bread, Hotdog, Cupcake.
- **-3 points (Crash):** Hitting a wall or a closed door.
- **+10 points (Finish line):** Getting to the present's finish line.
-

Game Over Conditions: A round ends when:

1. The vehicle crashes into a wall.
2. The vehicle successfully reaches the finish line.



<u>Pop Quiz</u>

Based on the Scoring Guide, which of these actions will result in the highest point deduction?

- ◯ Missing a piece of fruit
- ◯ Collecting junk food like a hotdog
- ◯ Crashing into a wall or a closed door
- ◯ Collecting a piece of fruit

When does a round of the game officially end?

- ◯ After the robot collects 10 items
- ◯ When the robot reaches the finish line OR crashes
- ◯ When the player presses the 'Stop' button
- ◯ Only when the robot crashes into a wall

Turorial

<u>-------- Practice time - Getting familiar with the Environment --------</u>

To get a better understanding we will let you control the car and play the game for 4 rounds.
Control Keys:
Left arrow (⟵) Move the car 1 step left
Right arrow (⟶)Move the car 1 step right
Space - Throw a key

You must paste that completion number to continue the survey.
Have fun!

A short video showing the game:

0:00 / 0:05

<u>Important Tip:</u> Allow for a slight delay between moves. Avoid rapidly pressing the arrow keys

The following link will open the game in a new browser tab - **link**

Paste your confirmation code here:

You are now ready to proceed to the **agent training part**!

Good luck!

Part 2: Feedback explanations

<u>Main Task - AI Agent Assistance</u>

Read the instructions very carefully, and move to the task.

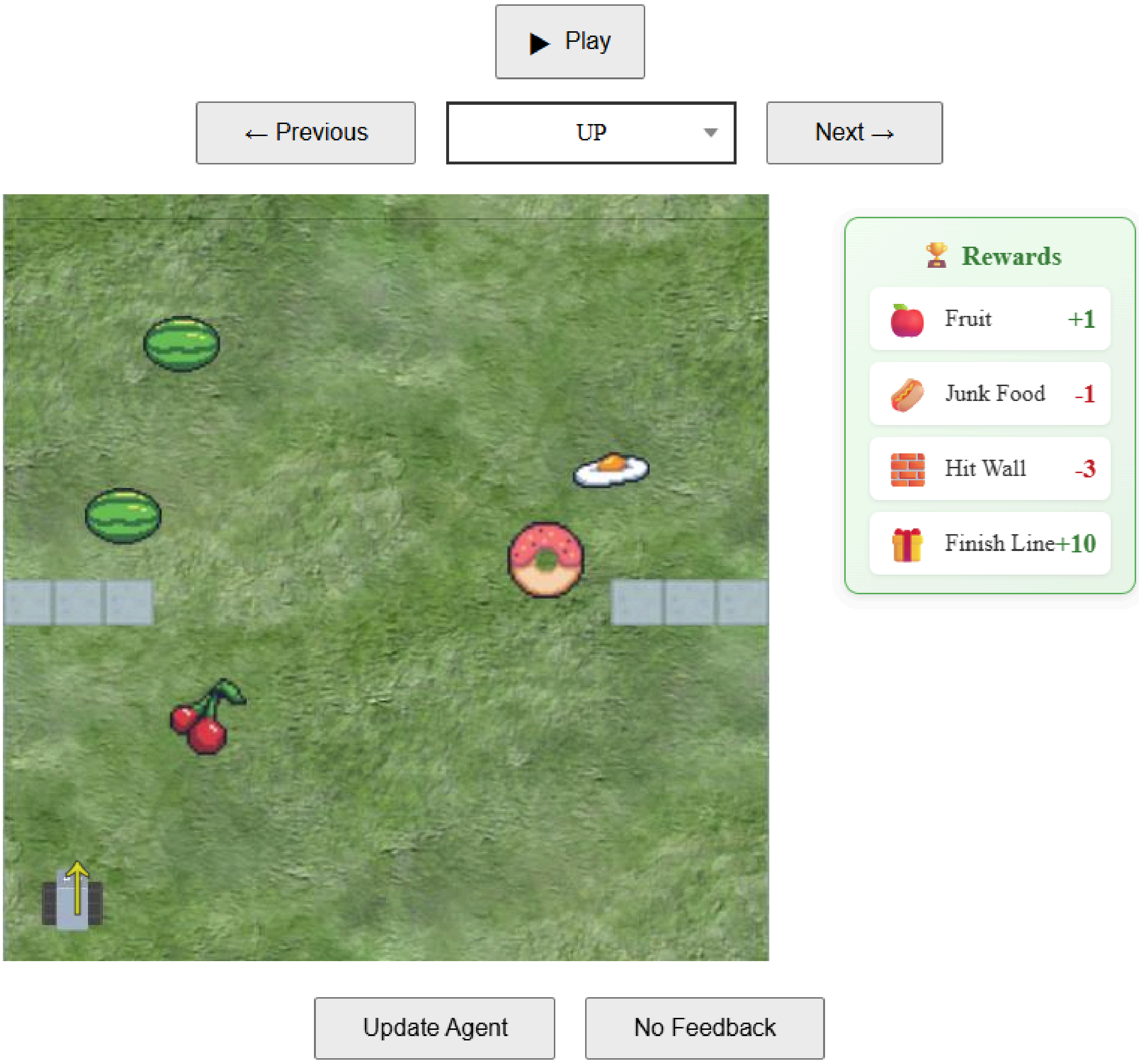
<u>Instructions:</u>
In this section, you will watch an AI agent play the game for **5 rounds**. The agent is currently imperfect—it will make mistakes and choose suboptimal actions.
Your Goal: Your job is to train the agent. **The better the agent performs by the end of the 5 rounds, the higher your bonus reward will be.**

How it Works:

1. **Watch:** In each round, you will first watch the agent play.
2. **Review:** Afterward, you will see a replay where you can step through the agent's moves.
3. **Correct:** You will identify mistakes and provide the correct action for each one.

   <u>Important</u>: You are reviewing the agent's latest path through the game. Your feedback applies only to this specific path. You may suggest better choices at particular moments, but this will not create a new path.

<u>Review Page:</u>

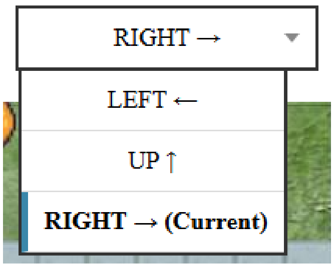


<u>**Navigate the Replay:**</u> Use the **Previous Action** and **Next Action** buttons to step through the path the agent took, one move at a time.
<u>**Make a Correction:**</u>
When you spot a suboptimal move:

- Click the action label (e.g., where it says "Right").
- Select the better alternative from the list.
  After selecting a correction, the updated action **highlighted in blue.**

- You can provide a correction **for each move the agent made**. However, we recommend no more than 10 corrections per round.

Note: when submitting multiple corrections for the same state, only your final selection will be saved.



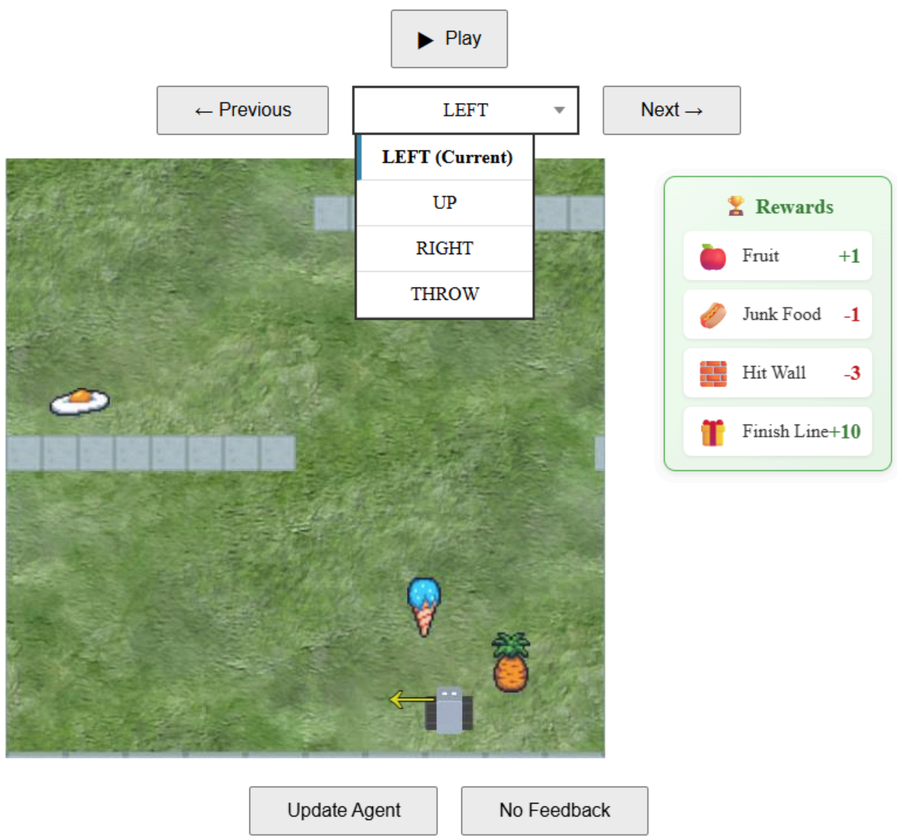
After Completing All Corrections Submit Your Feedback

- **Update Agent:** After submitting **all corrections** for this round, press the ***Update Agent*** button to update the agent's strategy for the next round.
- **No Feedback:** If the agent played perfectly and made no mistakes, click this to move to the next round without changes.

A short video demonstrating the feedback process:

0:00 / 1:00

Pop Quiz:



In the image above, what is the best feedback action to give the agent in that position?

- ◯ The current action (Left) is the best, no feedback is needed
- ◯ Right
- ◯ Up
- ◯ Throw

Main Task Link

Main Task
The game may take a few seconds to load and start.

The following link will open the game in a new browser tab - **link**

to continue, enter below the confirmation number you got at the end of the task:

Evaluate the agent

	Much worse	Worse	Somewhat worst	Same	Somewhat better	Better	Much better
How would you evaluate the agent's performance relative to yours in the game?	○	○	○	○	○	○	○

Demonstration Explanation - Same

Compare & Select the Better Agent

Updating an AI is complex. Sometimes your feedback improves the agent, but other times it might cause new, unintended behaviors.

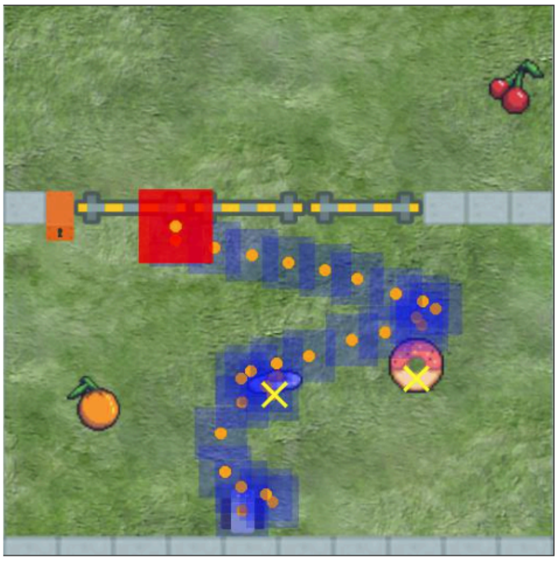
Your Task: We will show you a side-by-side visualization of the Previous Agent and the Updated Agent playing the same level.
You must decide which version is better to continue with — either accept the update and continue with the Updated Agent,
or reject the update and continue with the Previous Agent.

How to Read the visualization - Refer to the example image below

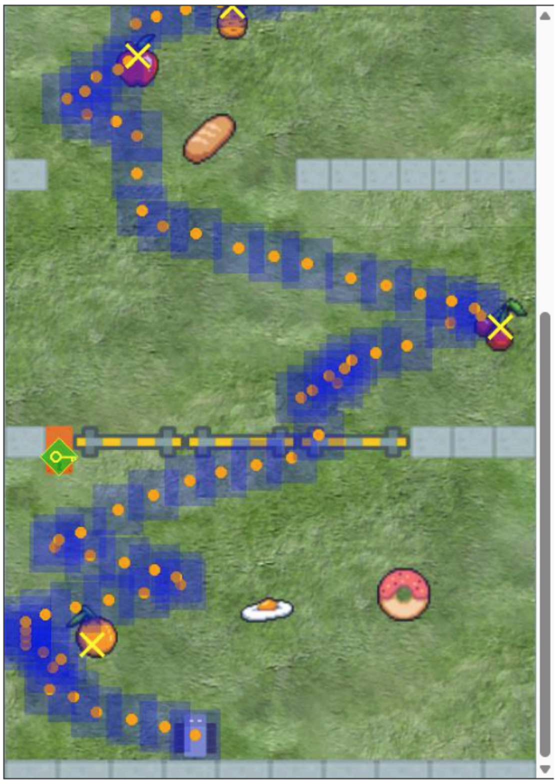
- **Blue trail** – The path the car traveled
- **Yellow X** – A food item was collected
- **Red block** – The car crashed into a wall or door (Game Over)
- **Green key** – A door was opened using a key

Compare Agent Behaviors

Previous Agent



Updated Agent



Use Previous Agent   Use Updated Agent

Decision:
If the new agent learned from your feedback and performed better, click Use ***Updated Agent***.
If the new agent got worse, click Use ***Previous Agent*** to discard the changes, and move to the next round.

After each update, you will be asked to evaluate how good the chosen agent is in general on a scale from 1 (very poor) to 7(excellent).

Rate the Chosen Agent

Please rate the chosen agent’s overall performance
(1 = very poor, 7 = excellent)

1 2 3 4 5 6 7

A short video demonstrating the comparison process:

0:00 / 0:36

Demonstration Explanation

Compare & Select the Better Agent

Updating an AI is complex. Sometimes your feedback improves the agent, but other times it might cause new, unintended behaviors.

Your Task: We will show you a side-by-side visualization of the Previous Agent and the Updated Agent playing the same level.
This level may be different from the one used for feedback.
You must decide which version is better to continue with — either accept the update and continue with the Updated Agent,
or reject the update and continue with the Previous Agent.

How to Read the visualization - Refer to the example image below

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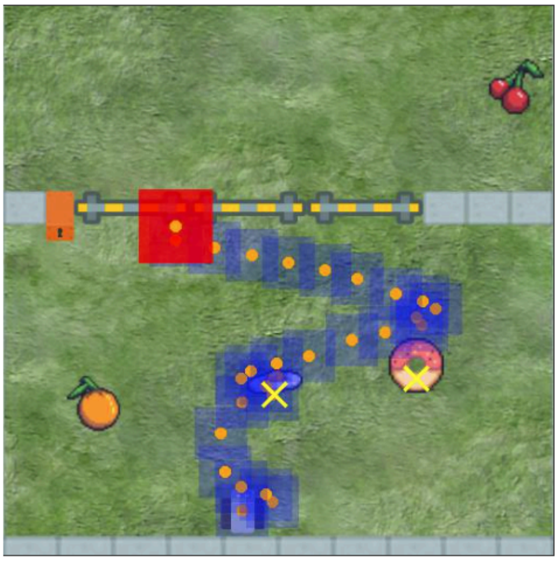
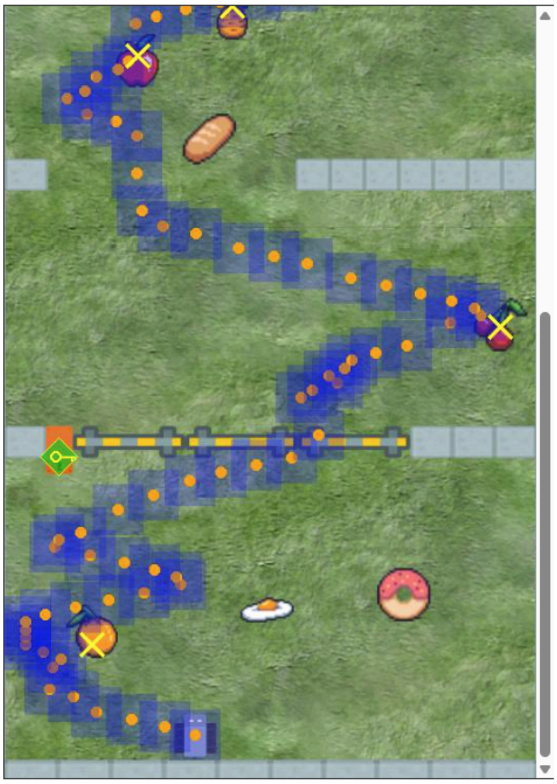


Updated Agent



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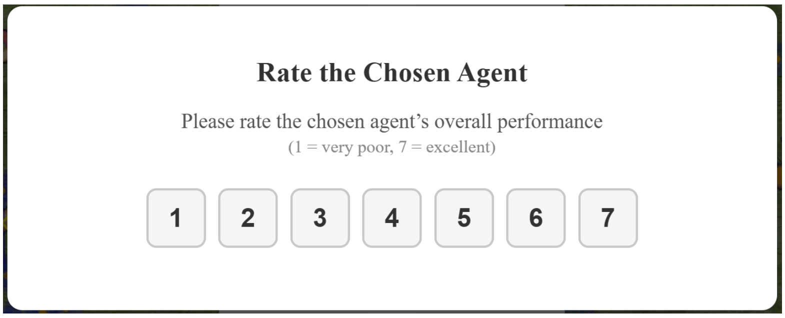

A short video demonstrating the comparison process:

0:00 / 0:36

System Satisfaction

Answer the following questions based on your opinion of the system.

	Strongly disagree	Disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Agree	Str a
The explanations of the agent’s paths were clear and easy to understand	○	○	○	○	○	○	
This is an attention check question, please select 'Strongly agree'	○	○	○	○	○	○	
The explanations felt too detailed and overwhelming	○	○	○	○	○	○	
The explanations helped me see how my feedback changed the agent’s behavior	○	○	○	○	○	○	
The visual demonstrations of policy changes helped me understand the differences between agents	○	○	○	○	○	○	
My feedback helped improve the agent's performance	○	○	○	○	○	○	

Part 3: Trust the Agent

Your bonus will depend on the success of **one final game round**.
In this round, you may choose who controls the vehicle:

- **The final trained AI agent**, which you helped train, or
- **You**, by playing the game yourself.

Please select your preferred option below.

○ I will play myself

○ I will let the AI agent play

Please briefly explain your choice:

Part3 - Play Yourself

Be ready, there is no second chance.
The following link will open the game in a new browser tab - **link**
Paste your confirmation code here:

Demographics

Finally, some background details:

Age:

Gender identity:

Male	Female	Non-Binary	Prefer not to say
◯	◯	◯	◯

In which country do you reside?

- ◯ United states
- ◯ United Kingdom
- ◯ Australia
- ◯ Canada
- ◯ Other

Years of education:

- ◯ Less than 8 years
- ◯ 8-12 years
- ◯ Bachelor level academic education
- ◯ Master level academic education
- ◯ PhD, MD, or other doctoral degree
- ◯ Other:

How familiar are you with Artificial Intelligence?

- ◯ Not at all familiar
- ◯ Slightly familiar (basic awareness, e.g., from news or media)
- ◯ Somewhat familiar (some knowledge, e.g., from a course or self-study)
- ◯ Moderately familiar (practical experience, e.g., using AI-powered tools or assistants)
- ◯ Very familiar (frequent user or hobbyist, e.g., building AI models or applications)
- ◯ Extremely familiar (professional experience, e.g., working in an AI-related field or industry)
- ◯ Expert level (advanced expertise, e.g., conducting research, developing new AI techniques or technologies)

Please state your agreement with the following sentences **regarding Artificial Intelligence (AI)**:

	1. Strongly disagree	2	3	4. Neither agree nor disagree	5	6	7. Strongly agree
I believe AI will improve my life.	◯	◯	◯	◯	◯	◯	◯
I believe that AI will improve my work.	◯	◯	◯	◯	◯	◯	◯
I think I will use AI technology in the future.	◯	◯	◯	◯	◯	◯	◯
I think AI technology is a threat to humans.	◯	◯	◯	◯	◯	◯	◯
I think AI technology is positive for humanity.	◯	◯	◯	◯	◯	◯	◯

Summary

Thank you for your participation.

Please share any thoughts or comments about this study in the space below.

Click the next button to complete the survey.